VaultX: A Lightweight Proof-of-Space Consensus System

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Motivation & Background

Proof-of-Work (PoW) secures Bitcoin by solving hash puzzles, burning electricity at national scale. An ASIC miner costs **$37K–$77K in electricity** over 5 years at $0.15/kWh.

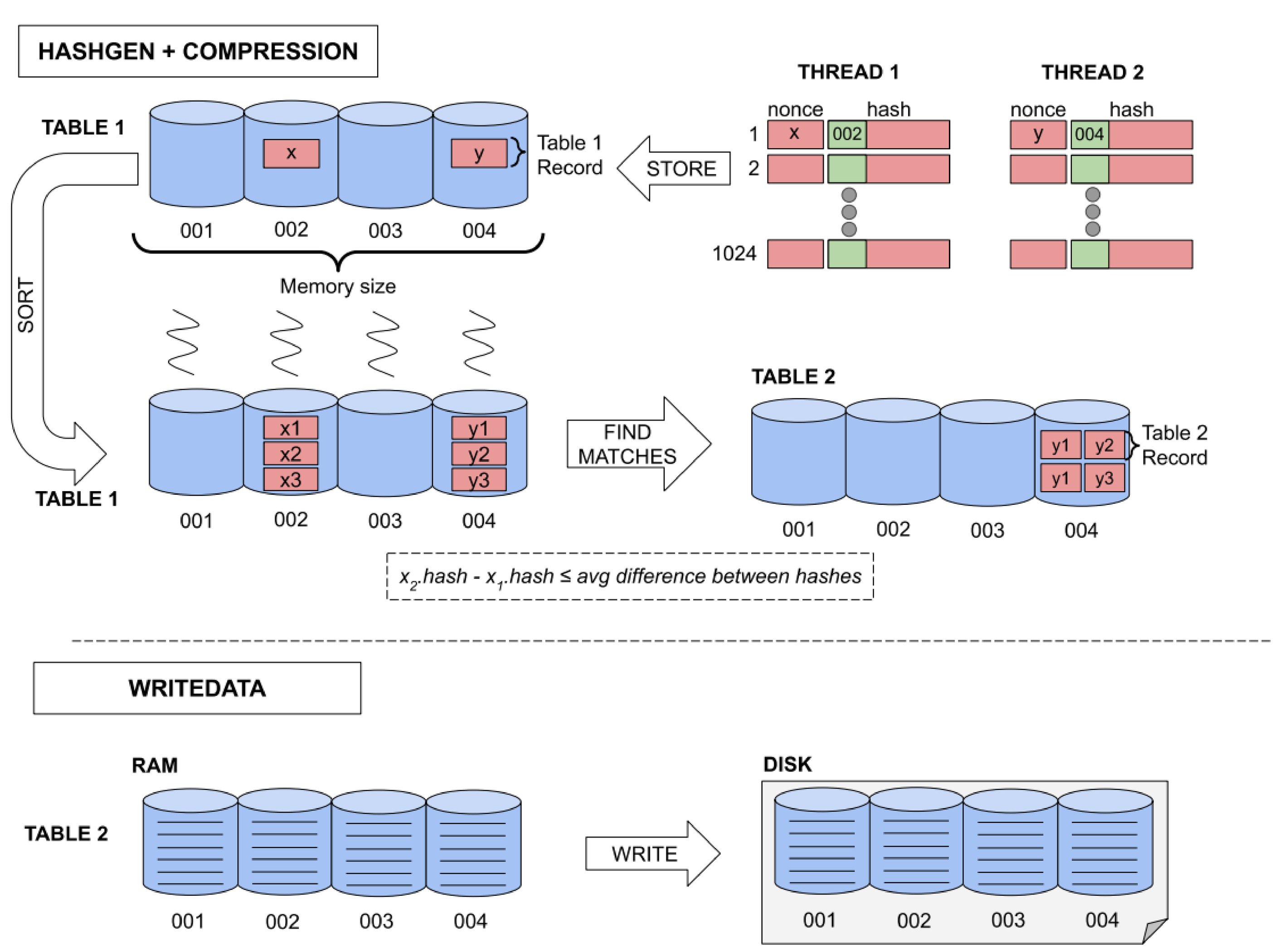
Proof-of-Space (PoSp) replaces computation with stored cryptographic data. Miners prove possession to earn rewards — no continuous energy burn.

Chia's limitations:

- 7-table SHA-256 pipeline — slow & complex
- Requires up to **416 GB RAM** (Bladebit) or GPU
- 101 GB plots vs. **32 GB VaultX vaults**
- Heavy SSD wear from multi-pass pipeline

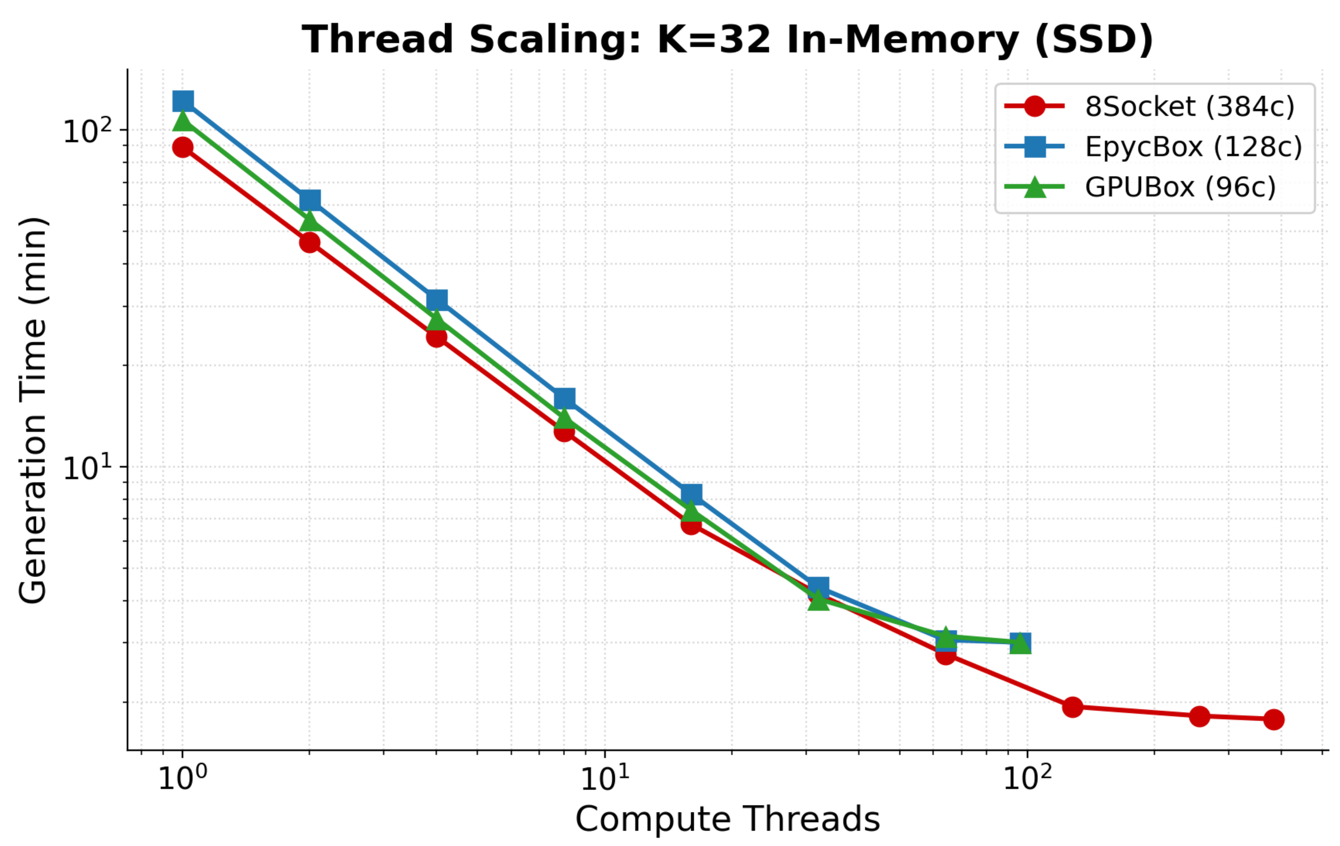
VaultX is a clean-slate PoSp: faster, smaller, simpler. Runs from Raspberry Pi to 192-core servers.

VaultX Architecture



(1) Key Derivation: BLAKE3 derives vault key. **(2) Table 1:** 2^K nonces hashed into 2^{24} buckets. **(3) Table 2:** Nonce pairs (i, j) satisfying hash-distance constraint stored as 8-byte records. **(4) Output:** Flat file enabling $O(1)$ lookup.

Thread Scaling (K=32, In-Memory, SSD)

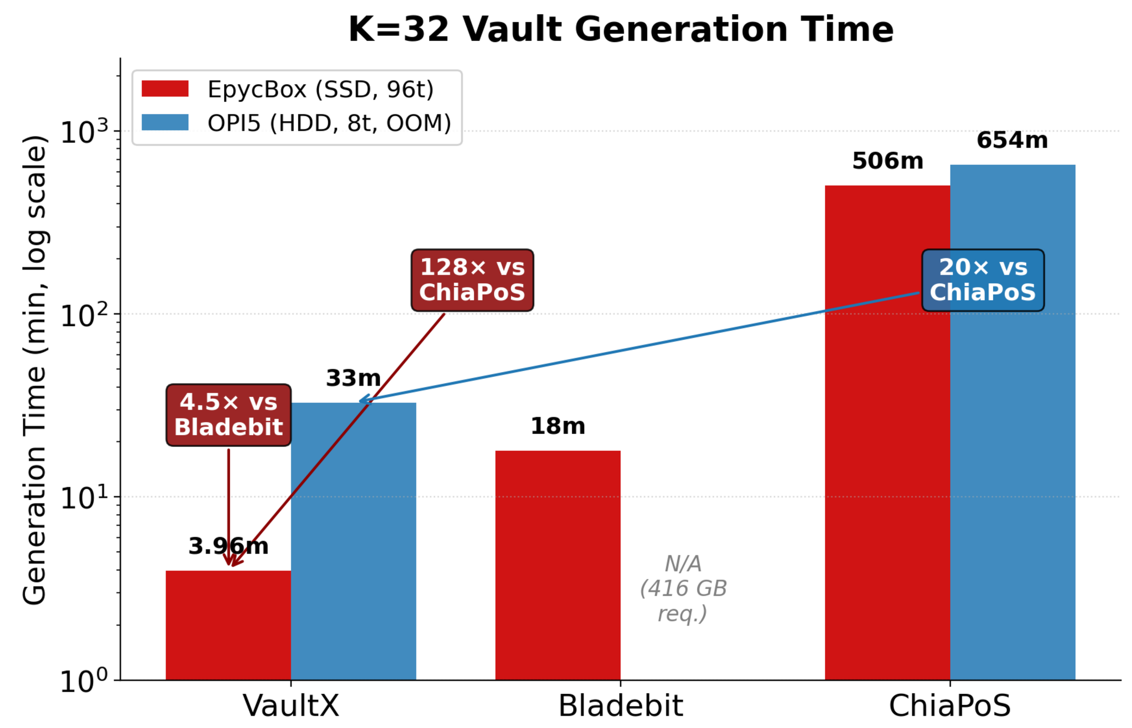


Near-linear scaling to 96–128 threads (I/O saturation). 8Socket: **1.74 min at 384 threads** (50× speedup).

K32 Vault Generation Performance

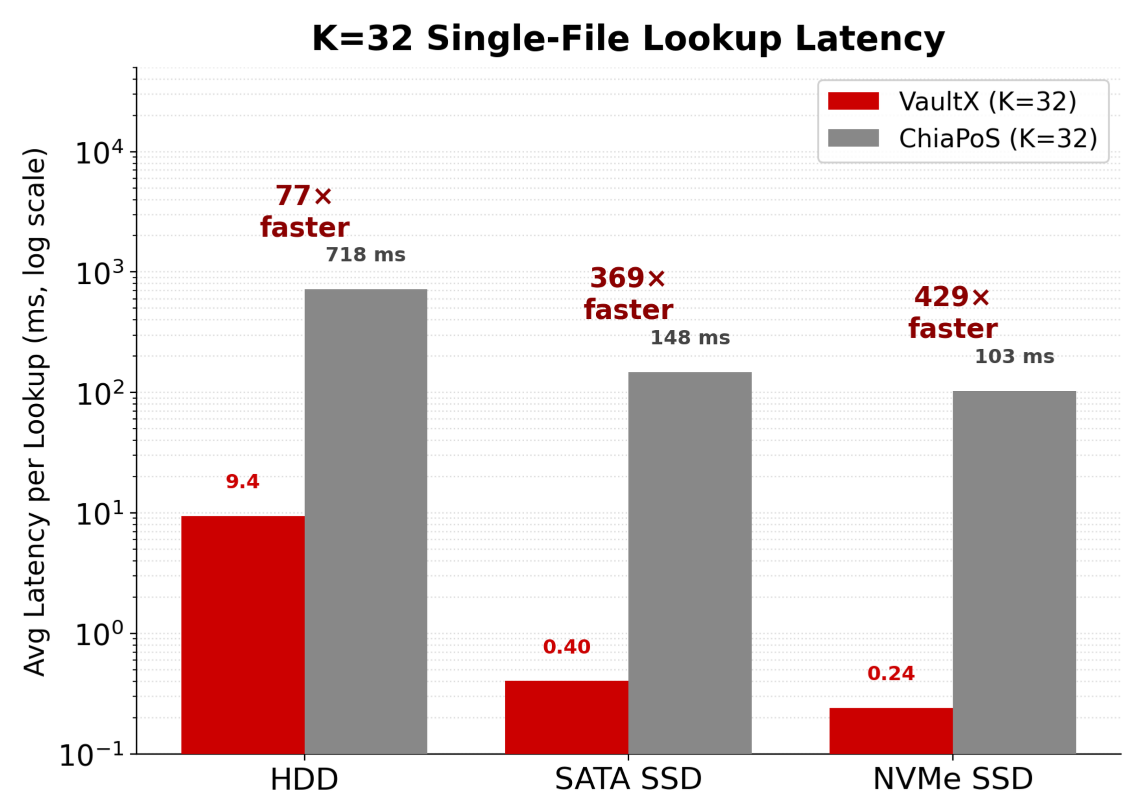
Machine	Cores	RAM	K=32 Time
8Socket	384	770 GB	1.74 m
ThunderX2	224	124 GB	5.95 m
EpycBox	128	192 GB	3.96 m
GPUBox	96	384 GB	3.01 m
ThunderX2	96	64 GB	6.2 m
NVMeBox	64	187 GB	2.82 m
Athena	48	64 GB	5.76 m
Torus	32	64 GB	6.28 m
OPI5	8	32 GB	32 m
RPI5	4	8 GB	74 m

11 machines tested across HDD, SATA SSD, NVMe, NFS, Ceph.



On **EpycBox**: VaultX K=32 in **3.96 min** — **4.5× faster** than Bladebit (17.9 m), **128× faster** than ChiaPoS (8 h 26 m), using only 50 GB RAM. On **OPI5**: **20× faster** than ChiaPoS.

Lookup Latency: VaultX vs. ChiaPoS



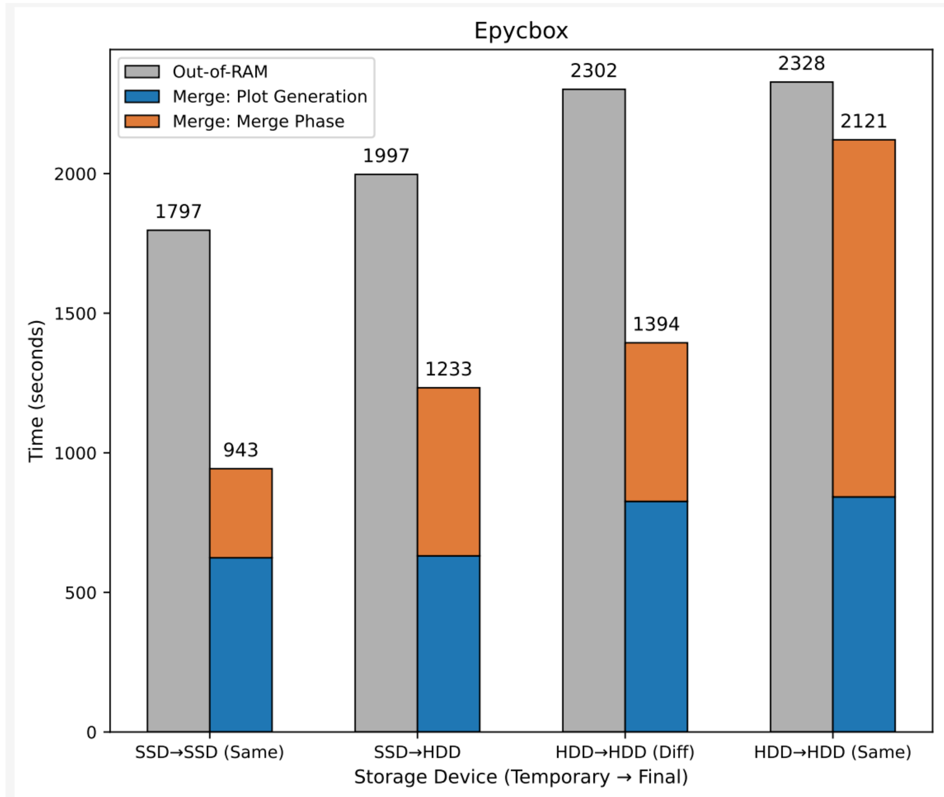
VaultX: $O(1)$ bucket + one seek. ChiaPoS: 7 tables + random seeks. **429× faster** on NVMe, **369×** on SATA SSD, **77×** on HDD.

Energy Consumption

Electricity cost per K=32 plot ($0.15/kWh):

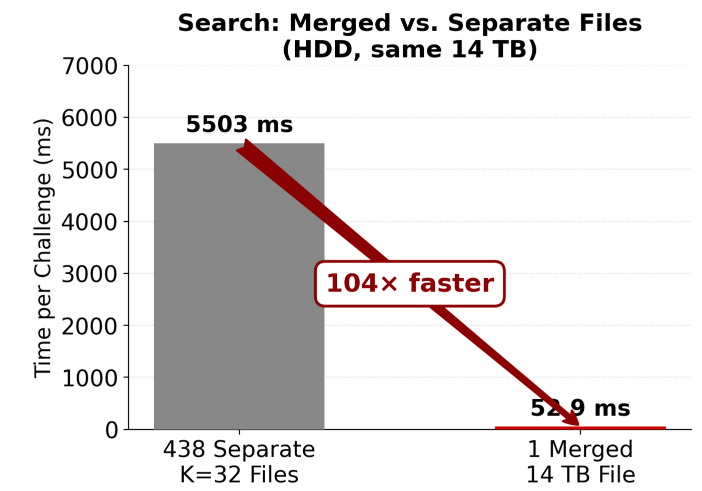
Machine	VaultX	Bladebit	ChiaPoS
OPI5	$0.002	—	$0.030
GPUbox	$0.006	$0.029	$0.833
EpycBox	$0.010	$0.041	$1.160
8Socket	$0.010	$0.063	$1.790

Merge-Based Large Vaults



Improving performance for K33 to K40 vault generation through the merge optimization.

Merged-File Search



438 separate K=32 files: 5503 ms (438 seeks). Merged: 53 ms (1 seek) — **104× speedup**. One-time merge; benefit per consensus round.

Key Contributions

1. **Two-table MEMOization** replaces Chia's 7-table pipeline.
2. **68% smaller vaults:** 32 GB (K=32) vs. 101 GB.
3. **4.5–128×** faster generation.
4. **BLAKE3 hashing:** 4–8× faster via SIMD. No GPU required.
5. **Near-linear OpenMP scaling** from **4-core Pi** to 192-core HPC server.
6. **Anti-Hellman design:** prevents time–memory attacks.
7. **Merge-based large vaults** (K=33–40) for memory-constrained machines.
8. **77–429×** lower lookup latency.
9. **17–179× lower electricity cost** than ChiaPoS.

References

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